

System & Operational Requirements Specification

Precision Hydro-Sensing Agricultural Platform with Closed-Loop Valve Actuation

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Repository:	https://github.com/projectvault/smart-irrigation-iot.git
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1. Specification Scope

This Technical Requirements Document establishes the operational specifications and system constraints for the Precision Hydro-Sensing Agricultural telemetry unit and valve control subsystem.

2. Operational Requirements

- REQ-A1 (Ground Metric Ingestion): Sensing probes must measure ground hydration and thermal parameters at 15-second cycles.
- REQ-A2 (Automated Discharge Control): The platform must energize hydraulic discharge relays whenever moisture index falls below 35%.
- REQ-A3 (Wireless Broker Publication): Ingested telemetry must be formatted in JSON and transmitted to the wireless broker topic 'agri/zone1/metrics'.
- REQ-A4 (Administrative Direct Override): The web console must furnish authorized operators with manual override controls for hydraulic lines.
- REQ-A5 (Emergency Timeout Protection): Discharge valves must automatically de-energize after 300 seconds if telemetry reception is interrupted.

3. System Quality Attributes

- Resilience: The sensing apparatus must maintain continuous field operation with minimal maintenance.
- Energy Efficiency: Micro-nodes must leverage low-power deep sleep cycles between telemetry transmissions.
- Command Responsiveness: Valve trigger commands must achieve execution within 1500 milliseconds.

4. Physical & Protocol Specifications

- Microcontroller: 32-bit Dual-Core Wireless SoC @ 240MHz.
- Communication: MQTT over TLS v1.3.
- Persistence Layer: Relational Time-Series Database.